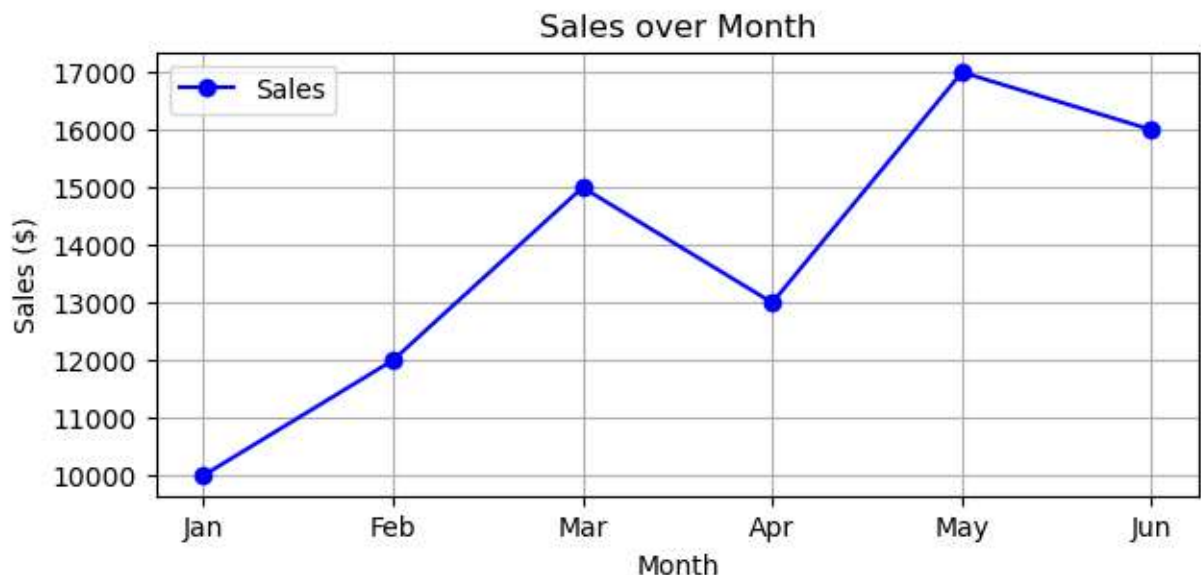


```
In [1]: import pandas as pd
data={
    'Month': ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun'],
    'Sales': [10000, 12000, 15000, 13000, 17000, 16000],
    'Profit': [2000, 3000, 4000, 2500, 3500, 3000]}
df = pd.DataFrame(data)
print(df)
```

	Month	Sales	Profit
0	Jan	10000	2000
1	Feb	12000	3000
2	Mar	15000	4000
3	Apr	13000	2500
4	May	17000	3500
5	Jun	16000	3000

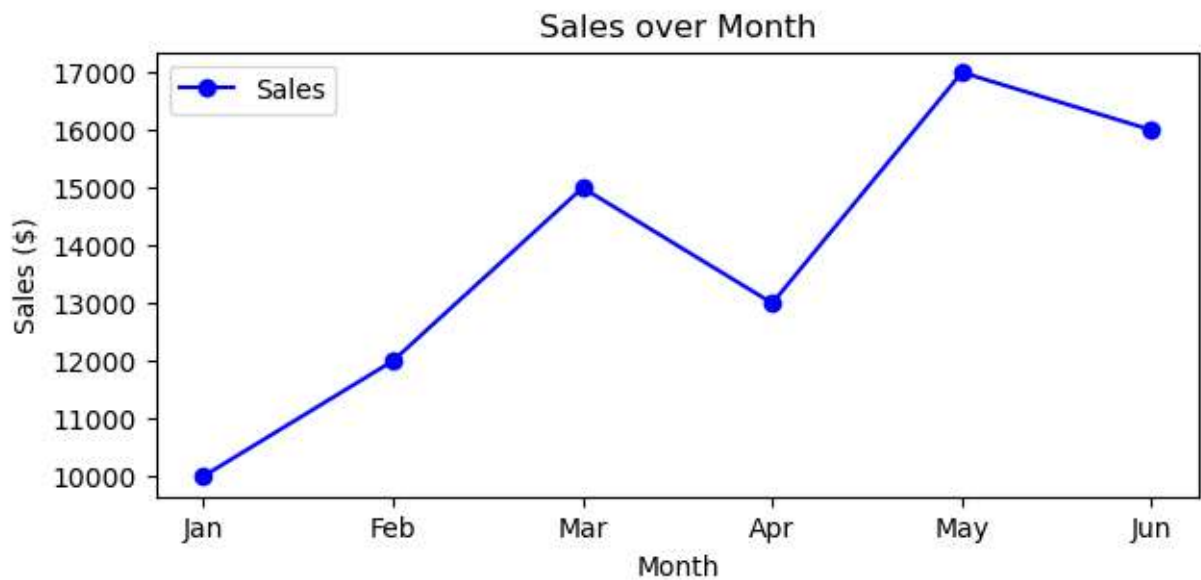
Line plot over monthly Sales

```
In [2]: import matplotlib.pyplot as plt
plt.figure(figsize=(7,3))
plt.plot(df['Month'],df['Sales'],color='blue',marker='o',ls='-',label='Sales')
plt.title('Sales over Month')
plt.xlabel('Month')
plt.ylabel('Sales ($)')
plt.grid(True)
plt.legend()
plt.show()
```



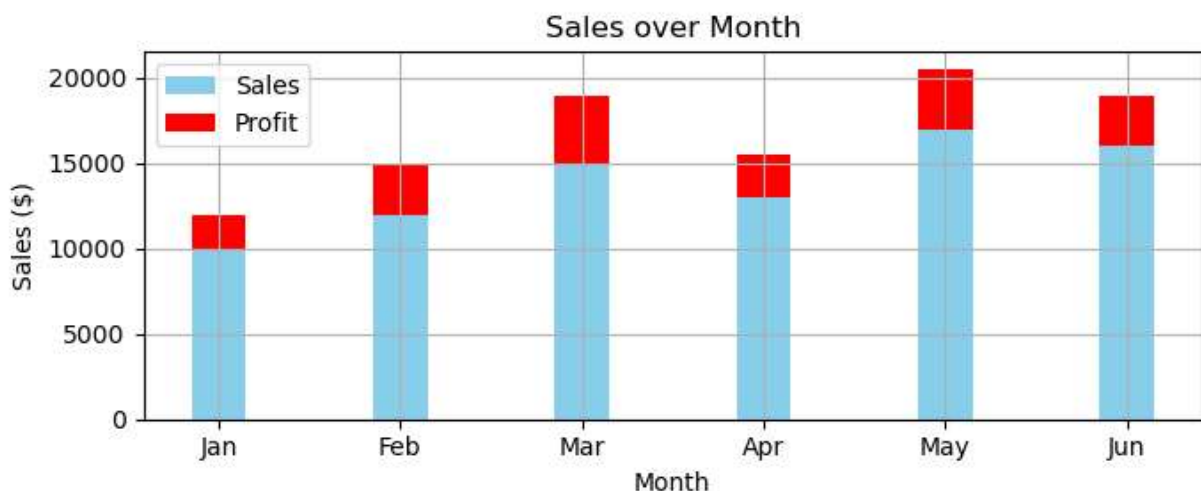
```
In [3]: import matplotlib.pyplot as plt
plt.figure(figsize=(7,3))
plt.plot(df['Month'],df['Sales'],color='blue',marker='o',ls='-',label='Sales')
plt.title('Sales over Month')
plt.xlabel('Month')
plt.ylabel('Sales ($)')
```

```
plt.grid(False)
plt.legend()
plt.show()
```



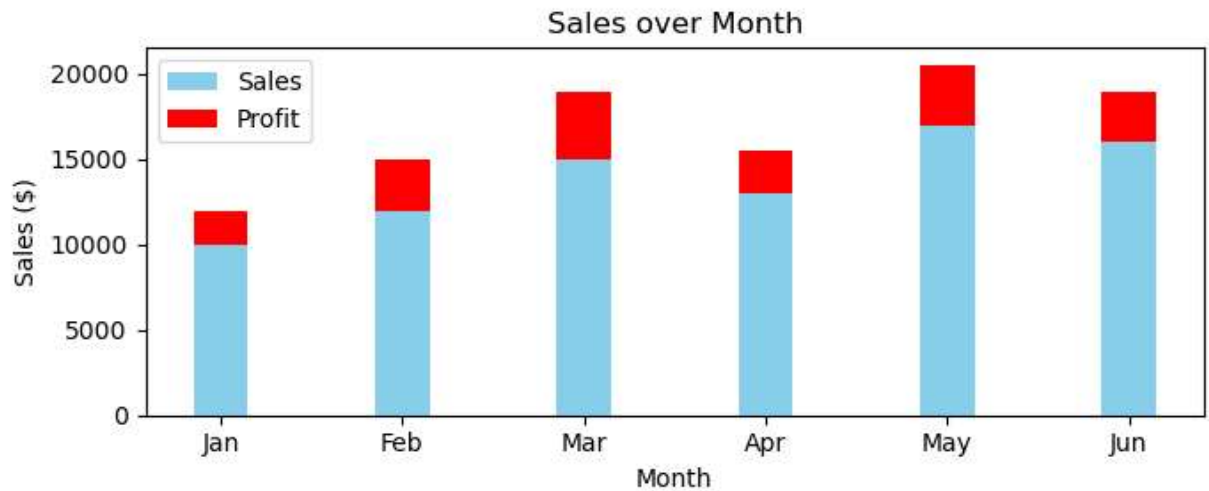
BarPlot between Month vs Profit

```
In [4]: plt.figure(figsize=(7,3))
width = 0.3
plt.bar(df['Month'],df['Sales'],width=width,color='skyblue',label='Sales')
plt.bar(df['Month'],df['Profit'],width=width,color='red',label='Profit',bottom=df['Sales'])
plt.title('Sales over Month')
plt.xlabel('Month')
plt.ylabel('Sales ($)')
plt.grid(True)
plt.legend()
plt.tight_layout()
plt.show()
```



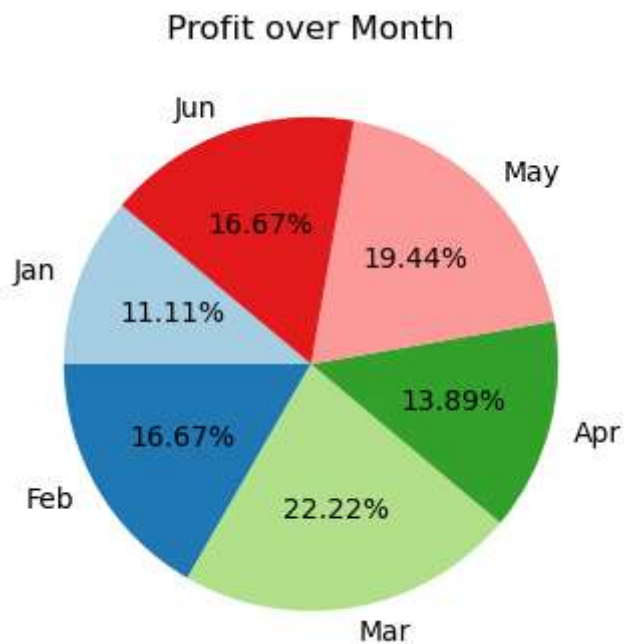
```
In [5]: plt.figure(figsize=(7,3))
width = 0.3
```

```
plt.bar(df['Month'],df['Sales'],width=width,color='skyblue',label='Sales')
plt.bar(df['Month'],df['Profit'],width=width,color='red',label='Profit',bottom=df['Sales'])
plt.title('Sales over Month')
plt.xlabel('Month')
plt.ylabel('Sales ($)')
plt.grid(False)
plt.legend()
plt.tight_layout()
plt.show()
```



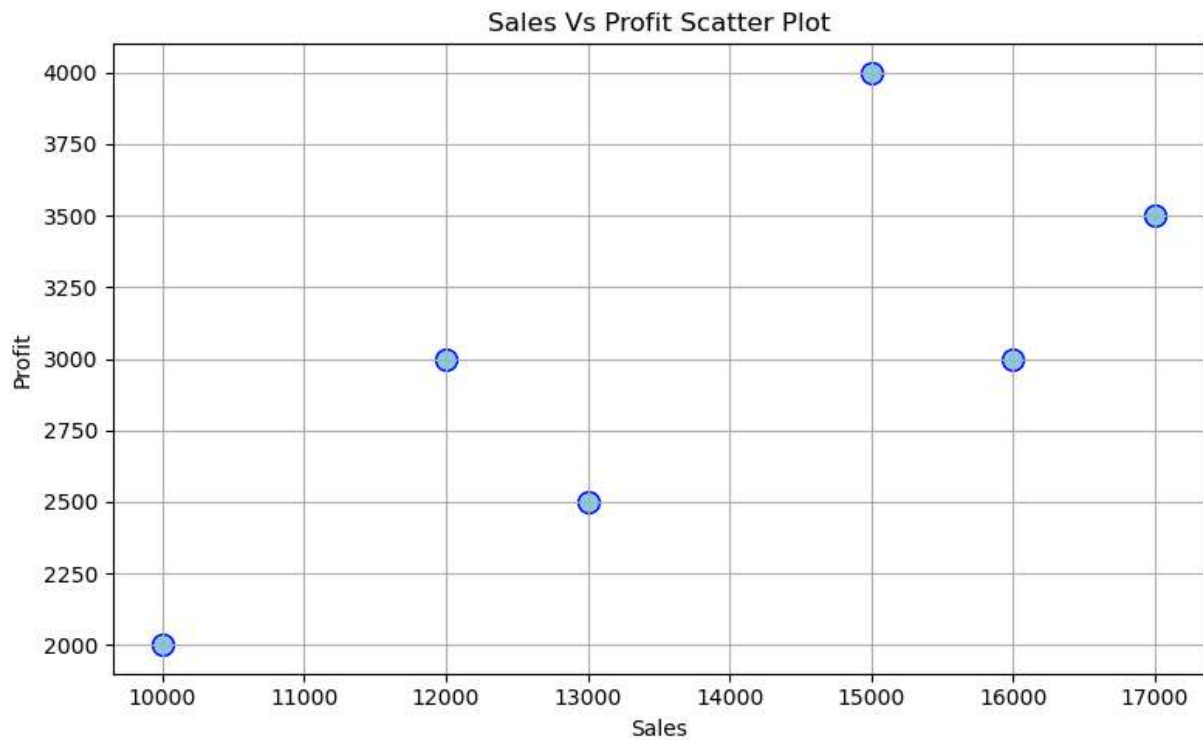
Pie Chart B/w Profit vs Month

```
In [6]: from enum import auto
plt.figure(figsize=(10,4))
plt.pie(df['Profit'], labels = df['Month'],autopct='%1.2f%%',startangle=140,colors=
plt.title('Profit over Month')
plt.show()
```



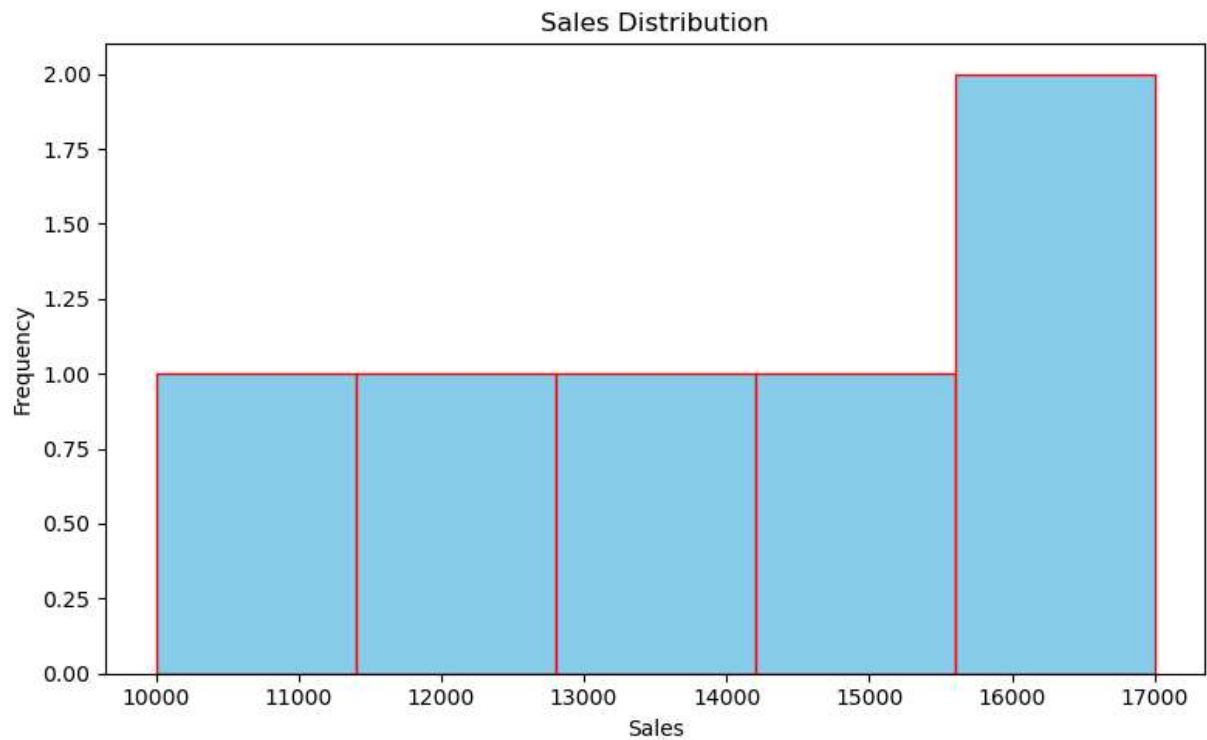
Scatter Plot

```
In [7]: plt.figure(figsize=(8,5))
plt.scatter(df['Sales'],df['Profit'],color='skyblue',s=100,edgecolors='blue')
plt.title('Sales Vs Profit Scatter Plot')
plt.xlabel('Sales')
plt.ylabel('Profit')
plt.tight_layout()
plt.grid(True)
plt.show()
```



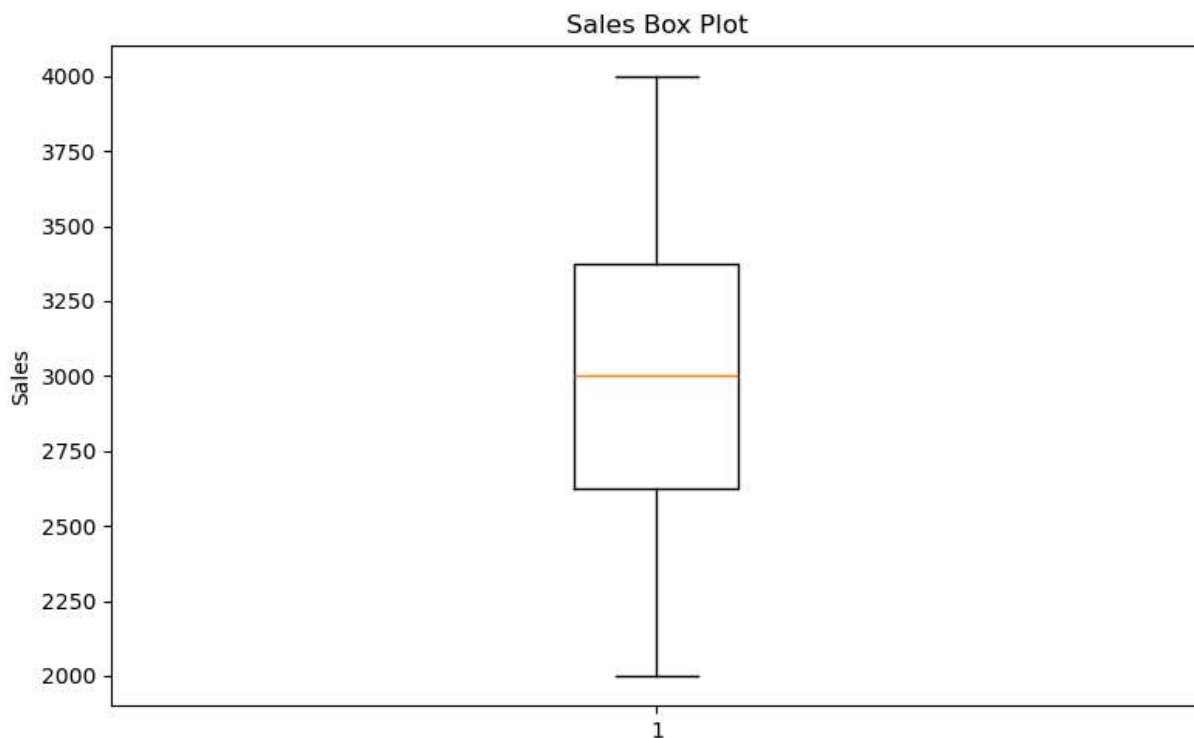
Histogram

```
In [8]: plt.figure(figsize=(8,5))
plt.hist(df['Sales'],bins=5,color='skyblue',edgecolor='red'),
plt.title('Sales Distribution')
plt.xlabel('Sales')
plt.ylabel('Frequency')
plt.tight_layout()
plt.show()
```

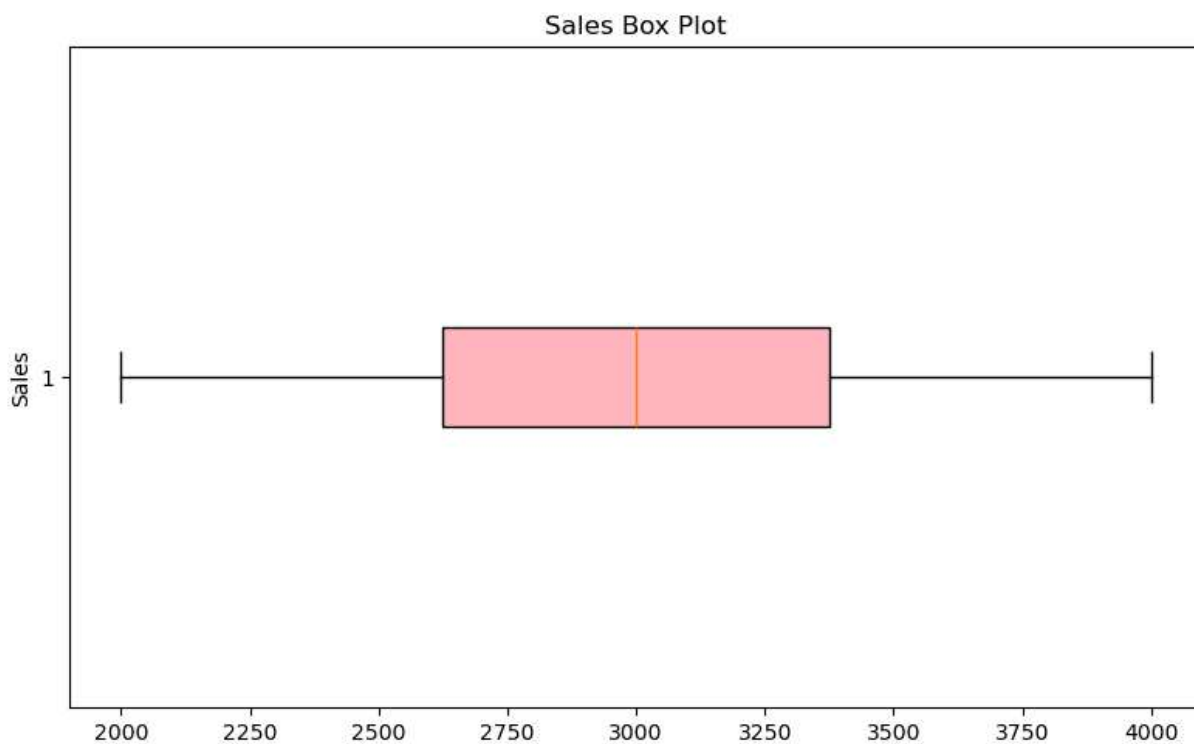


Box Plot

```
In [9]: plt.figure(figsize=(8,5))
plt.boxplot(df['Profit'])
plt.title('Sales Box Plot')
plt.ylabel('Sales')
plt.tight_layout()
plt.show()
```



```
In [10]: plt.figure(figsize=(8,5))
plt.boxplot(df['Profit'],vert=False,patch_artist=True,boxprops=dict(facecolor='lightcoral'))
plt.title('Sales Box Plot')
plt.ylabel('Sales')
plt.tight_layout()
plt.show()
```



```
In [20]: !pip install gradio
```

Requirement already satisfied: gradio in c:\users\suras\anaconda2\lib\site-packages (5.37.0)

Requirement already satisfied: aiofiles<25.0,>=22.0 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (24.1.0)

Requirement already satisfied: anyio<5.0,>=3.0 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (4.2.0)

Requirement already satisfied: brotli>=1.1.0 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (1.1.0)

Requirement already satisfied: fastapi<1.0,>=0.115.2 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (0.116.1)

Requirement already satisfied: ffmpeg in c:\users\suras\anaconda2\lib\site-packages (from gradio) (0.6.0)

Requirement already satisfied: gradio-client==1.10.4 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (1.10.4)

Requirement already satisfied: groovy~=0.1 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (0.1.2)

Requirement already satisfied: httpx<1.0,>=0.24.1 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (0.27.0)

Requirement already satisfied: huggingface-hub>=0.28.1 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (0.33.4)

Requirement already satisfied: jinja2<4.0 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (3.1.4)

Requirement already satisfied: markupsafe<4.0,>=2.0 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (2.1.3)

Requirement already satisfied: numpy<3.0,>=1.0 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (1.26.4)

Requirement already satisfied: orjson~=3.0 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (3.10.18)

Requirement already satisfied: packaging in c:\users\suras\anaconda2\lib\site-packages (from gradio) (24.1)

Requirement already satisfied: pandas<3.0,>=1.0 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (2.2.2)

Requirement already satisfied: pillow<12.0,>=8.0 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (10.4.0)

Requirement already satisfied: pydantic<2.12,>=2.0 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (2.8.2)

Requirement already satisfied: pydub in c:\users\suras\anaconda2\lib\site-packages (from gradio) (0.25.1)

Requirement already satisfied: python-multipart>=0.0.18 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (0.0.20)

Requirement already satisfied: pyyaml<7.0,>=5.0 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (6.0.1)

Requirement already satisfied: ruff>=0.9.3 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (0.12.3)

Requirement already satisfied: safehttpx<0.2.0,>=0.1.6 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (0.1.6)

Requirement already satisfied: semantic-version~=2.0 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (2.10.0)

Requirement already satisfied: starlette<1.0,>=0.40.0 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (0.47.1)

Requirement already satisfied: tomlkit<0.14.0,>=0.12.0 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (0.13.3)

Requirement already satisfied: typer<1.0,>=0.12 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (0.16.0)

Requirement already satisfied: typing-extensions~=4.0 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (4.11.0)

Requirement already satisfied: uvicorn>=0.14.0 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (0.35.0)

Requirement already satisfied: fsspec in c:\users\suras\anaconda2\lib\site-packages (from gradio-client==1.10.4->gradio) (2024.6.1)

Requirement already satisfied: websockets<16.0,>=10.0 in c:\users\suras\anaconda2\lib\site-packages (from gradio-client==1.10.4->gradio) (10.4)

Requirement already satisfied: idna>=2.8 in c:\users\suras\anaconda2\lib\site-packages (from anyio<5.0,>=3.0->gradio) (3.7)

Requirement already satisfied: sniffio>=1.1 in c:\users\suras\anaconda2\lib\site-packages (from anyio<5.0,>=3.0->gradio) (1.3.0)

Requirement already satisfied: certifi in c:\users\suras\anaconda2\lib\site-packages (from httpx<1.0,>=0.24.1->gradio) (2024.8.30)

Requirement already satisfied: httpcore==1.* in c:\users\suras\anaconda2\lib\site-packages (from httpx<1.0,>=0.24.1->gradio) (1.0.2)

Requirement already satisfied: h11<0.15,>=0.13 in c:\users\suras\anaconda2\lib\site-packages (from httpcore==1.*->httpx<1.0,>=0.24.1->gradio) (0.14.0)

Requirement already satisfied: filelock in c:\users\suras\anaconda2\lib\site-packages (from huggingface-hub>=0.28.1->gradio) (3.13.1)

Requirement already satisfied: requests in c:\users\suras\anaconda2\lib\site-packages (from huggingface-hub>=0.28.1->gradio) (2.32.3)

Requirement already satisfied: tqdm>=4.42.1 in c:\users\suras\anaconda2\lib\site-packages (from huggingface-hub>=0.28.1->gradio) (4.66.5)

Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\suras\anaconda2\lib\site-packages (from pandas<3.0,>=1.0->gradio) (2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in c:\users\suras\anaconda2\lib\site-packages (from pandas<3.0,>=1.0->gradio) (2024.1)

Requirement already satisfied: tzdata>=2022.7 in c:\users\suras\anaconda2\lib\site-packages (from pandas<3.0,>=1.0->gradio) (2023.3)

Requirement already satisfied: annotated-types>=0.4.0 in c:\users\suras\anaconda2\lib\site-packages (from pydantic<2.12,>=2.0->gradio) (0.6.0)

Requirement already satisfied: pydantic-core==2.20.1 in c:\users\suras\anaconda2\lib\site-packages (from pydantic<2.12,>=2.0->gradio) (2.20.1)

Requirement already satisfied: click>=8.0.0 in c:\users\suras\anaconda2\lib\site-packages (from typer<1.0,>=0.12->gradio) (8.1.7)

Requirement already satisfied: shellingham>=1.3.0 in c:\users\suras\anaconda2\lib\site-packages (from typer<1.0,>=0.12->gradio) (1.5.4)

Requirement already satisfied: rich>=10.11.0 in c:\users\suras\anaconda2\lib\site-packages (from typer<1.0,>=0.12->gradio) (13.7.1)

Requirement already satisfied: colorama in c:\users\suras\anaconda2\lib\site-packages (from click>=8.0.0->typer<1.0,>=0.12->gradio) (0.4.6)

Requirement already satisfied: six>=1.5 in c:\users\suras\anaconda2\lib\site-packages (from python-dateutil>=2.8.2->pandas<3.0,>=1.0->gradio) (1.16.0)

Requirement already satisfied: markdown-it-py>=2.2.0 in c:\users\suras\anaconda2\lib\site-packages (from rich>=10.11.0->typer<1.0,>=0.12->gradio) (2.2.0)

Requirement already satisfied: pygments<3.0.0,>=2.13.0 in c:\users\suras\anaconda2\lib\site-packages (from rich>=10.11.0->typer<1.0,>=0.12->gradio) (2.15.1)

Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\suras\anaconda2\lib\site-packages (from requests->huggingface-hub>=0.28.1->gradio) (3.3.2)

Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\suras\anaconda2\lib\site-packages (from requests->huggingface-hub>=0.28.1->gradio) (1.26.20)

Requirement already satisfied: mdurl~>=0.1 in c:\users\suras\anaconda2\lib\site-packages (from markdown-it-py>=2.2.0->rich>=10.11.0->typer<1.0,>=0.12->gradio) (0.1.0)

```
In [35]: import gradio as gr
import pandas as pd
import matplotlib.pyplot as plt
```

```

data={
    'Month':['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun'],
    'Sales':[10000,12000,15000,13000,17000,16000],
    'Profit':[2000,3000,4000,2500,3500,3000]}
df=pd.DataFrame(data)

def generate_plot(plot_type):
    fig=plt.figure(figsize=(7,3))
    if plot_type=='Line Plot':
        plt.plot(df['Month'],df['Sales'],color='blue',marker='o',ls='-',label='Sale
        plt.title('Sales over Month')
        plt.xlabel('Month')
        plt.ylabel('Sales ($)')
        plt.grid(False)
        plt.legend()

    elif plot_type == 'stacked bar chare':
        fig.set_size_inches(6,3)
        width=0.3
        plt.bar(df['Month'],df['Sales'],width=width,color='skyblue',label='Sales')
        plt.bar(df['Month'],df['Profit'],width=width,color='red',label='Profit',bot
        plt.title('Sales over Month')
        plt.xlabel('Month')
        plt.ylabel('Sales ($)')
        plt.grid(True)
        plt.legend()

    elif plot_type=='Pie chart':
        fig.set_size_inches(7,2)
        plt.figure(figsize=(10,4))
        plt.pie(df['Profit'], labels = df['Month'],autopct='%1.2f%%',startangle=140
        plt.title('Profit over Month')

    elif plot_type=='Scatter Plot':
        plt.scatter(df['Sales'],df['Profit'],color='skyblue',s=100,edgecolors='blue
        plt.title('Sales Vs Profit Scatter Plot')
        plt.xlabel('Sales')
        plt.ylabel('Profit')
        plt.grid(True)

    elif plot_type=='Histogram':
        plt.hist(df['Sales'],bins=5,color='purple',edgecolor='black')
        plt.title('Sales Distribution')
        plt.xlabel('Sales($')
        plt.ylabel('Frequency')

    elif plot_type=='Box Plot':
        plt.boxplot(df['Profit'],vert=False,patch_artist=True,boxprops=dict(facecol
        plt.title('Sales Box Plot')
        plt.xlabel('Profit ($)')

    plt.tight_layout()
    return fig

#gradio

```

```
demo= gr.Interface(  
    fn=generate_plot,  
    inputs=gr.Radio(['Line Plot','stacked bar chart','Pie Chart','Scatter Plot','Hi  
    label='choose plot type'),  
    outputs=gr.Plot(label='Visualization'),  
    title='Sales & Profit Visual Insight',  
    description='choose the type of visualize data')  
demo.launch()
```

* Running on local URL: <http://127.0.0.1:7864>

* To create a public link, set `share=True` in `launch()`.

Out[35]:

Using existing dataset file at: .gradio\flagged\dataset1.csv

In []: